# RIN RAY

rin.io/about · cray@uni-muenster.de · they/them
CRC Deformations-Geometry-Structure Postdoctoral Fellow, Universität Münster
*arithmetic at every chromatic height*

## ACADEMIC EMPLOYMENT

2023–Present — **CRC Postdoctoral Fellow**; Universität Münster; Arithmetic and Homotopy Theory Working Group, with Thomas Nikolaus and Christopher Deninger
- chromatic homotopy theory · Eisenstein series and categorical traces · genera and equivariant $K$-theory · zeta functions and special values · algebraic $K$-theory

## EDUCATION

2017–2023 — **Northwestern**; PhD in Mathematics; advised by Paul Goerss
- *A Geometric Model of Higher K-theories at Height $h=p^{k-1}(p-1)$ via Families of Ramified Curves*

2015–2017 — **UChicago**; Divisional Masters Program in Mathematics; advised by Peter May and Kazuya Kato
- *Calculating $\pi_*(tmf)$ at the prime 2*
- *Models of Formal Group Laws of Every Height*

2012–2013 — **George Mason University**; B.S. in Computational Physics (graduated winter 2013, at 16); advised by Dimitrios Papaconstantopoulos
- *Contextual Machine Learning through the Analysis and Chunking of Partially Translated Grade 2 Braille*
- *Predicting Superconductivity Transition Temperature via the Augmented Plane Wave Model*

2010–2012 — **Mary Baldwin College**; Program for the Exceptionally Gifted: Mathematics, Computer Science, and Physics; advised by John Ong

## RESEARCH

Nov 2026 — **Only the Gauge Grows: Moduli Stacks of $G$-Curves in Homotopy Theory at $h = p^{k-1}(p-1)$** on arXiv soon; Part III

Sept 2026 — **On the Chromatic Splitting Conjecture in Coheight 1**; joint with T. Barthel, L. Mann, A. Senger, T. Schlank, J. Weinstein, X. Zhou; coheight-one weak chromatic splitting at every height, for generic primes; before, known only at heights 1 and 2

Sept 2025 — V3 OCT 2026 — **Tiny and Unique: The Lubin–Tate Action at Height $p-1$ from the Minimal Curve with Its Symmetries** submitted: *AG&T*; arXiv:2509.23428

July 2025 — **Towards the $p = 3$ Kervaire Invariant Problem** submitted; joint with E. Belmont; the $E_2$-page for the homotopy fixed points spectral sequence computing $\pi_*(E_6{}^{hC_9})$; arXiv:2507.10157 · both our results and the method we proposed for the remaining case were used in a claimed resolution of the $p = 3$ Kervaire invariant problem (OpenAI preprint, Sept 2026)

July 2025 — V2 OCT 2026 — **Modeling Group Actions on Stacks (Especially the Lubin–Tate Action)** submitted: *Journal of Geometry and Physics*; arXiv:2507.00309

2022 — **Automorphisms of Abelian Varieties and Principal Polarizations** published; joint with D. Lee; *Rendiconti del Circolo Matematico di Palermo*, Series 2, vol. 71, pp. 483–494; arXiv:1811.07007

2020 — **Towards Directed Collapsibility** published; joint with R. Belton, R. Brooks, S. Ebli, L. Fajstrup, B. T. Fasy, N. Sanderson, E. Vidaurre; *Advances in Mathematical Sciences*, vol. 21, pp. 255–271; arXiv:1902.01039

2018 — **A Global Crystalline Period Map**; joint with M. Neaton and A. Pieper; from the Arizona Winter School; arXiv:1911.08615

## CURRENT PROJECTS

Scattering — **The Unramified Eisenstein Spectrum as a Categorical Trace**; solo

Echoes — **The λ-twisted $T(h+1)$-local Zeta Function**; joint with G. Angelini-Knoll

Echoes — **Toward Categorifying Symplectic $L$-functions and Reidemeister Torsion**; joint with A. Petrov and M. Volpe

Echoes — ***L*-functions and the Relative Image of $J$ in Moore Spectra**

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| Echoes | **All Bernoulli Numbers in Homotopy Theory Are Shifts**; joint with A. Mejia and N. Riggenbach |

## FELLOWSHIPS AND AWARDS

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| 2022 | **Gelfand Prize**, Northwestern University; for outstanding progress in one's own research and extraordinary contributions to the research goals of the graduate community |
| 2022 | **Nominee, Presidential Fellowship**, Northwestern University |
| 2017 | Awarded **NSF GRFP Fellowship** (103K + 36K) |
| 2014–2016 | Awarded **Thiel Fellowship** (100K); mathematics and assistive technology; mentor Edward Frenkel |

## TEACHING

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| 2025 | **Mathe trifft Kunst** (Math Meets Art), Mathematics Münster; organized all the children's activities, about 100 children: Escher drawings and wallpaper groups, dual polytopes, Sierpiński triangles |
| 2024 | **Feedback seminar**, Homotopy working group, Münster; assigned feedback-givers for every talk |
| 2018–2025 | **Speaker coaching**; coached every speaker in the seminars I organized, at Northwestern and in Münster |
| 2017–2023 | **Teaching Assistant**, Northwestern; discussion sections, quizzes and solutions: Intensive Linear Algebra and Multivariable Calculus (MATH 291, honors), Linear Algebra (MATH 240), Discrete Mathematics · evaluations (2021): 5.65/6 interest in teaching, 5.48/6 clarity |
| 2020–2023 | **Directed Reading Program**, Northwestern; weekly one-on-one reading with an undergraduate, ending in a talk to their peers |
| 2015–2017 | **Research Experience for Undergraduates**, University of Chicago; weekly lectures, and guiding each student through an individual research project |
| 2016–2017 | **Math Circles**, South Side of Chicago; weekly mathematics with children aged 6–12 |
| 2012 | **Teaching Assistant, College Algebra**, Mary Baldwin College |

## COMMUNITY AND OUTREACH

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| 2024 | **Queer in Math**; panelist at Queer in Math Day, MPI Leipzig |
| 2023 | **Set up a refugee program for PhD students**, Universität Münster; hosted PhD students for up to six months, waiving the usual processing time so they could leave at once |
| 2022 | **Advocate for Ukrainian students**, Northwestern; won incompletes and leniency for students on visas |
| 2021–2023 | **Causeway Bridge Program** and **TGS Diversity Peer Mentorship**, Northwestern; mentoring students from diverse backgrounds into and through graduate school |
| 2021– | **Founder, Mental Health Math Buddy Program**, Northwestern; pairs first- and second-year graduate students with older students for quarterly check-ins; ongoing |
| 2017–2023 | **Building Cities with Young Girls**, Evanston; weekly, to give working mothers in math time back |
| 2022 | **Founder and Treasurer, Graduate Life Appreciation Fund**, Northwestern |
| 2020–2023 | **Medical Foster**, Evanston Animal Shelter and Red Door Animal Shelter, Evanston; Tree House Humane Society, Chicago; cared at home for animals too sick or too young to stay in the shelter: special-needs adult cats and neonatal kittens |
| ongoing | ***Good Fibrations***, expository math blog with drawings, rin.io; and *re-search*, an interactive story about understanding something for the first time |
| 2014– | **Founding contributor, Topos House**, South San Francisco; a house for collaboration between mathematicians, biologists and artists; ongoing |

## SEMINARS AND CONFERENCES ORGANIZED

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| 2024 | **Young Topologists Meeting**, Universität Münster; organizing committee; drew the logo, designed the schedule with N. Pacheco-Tallaj |
| 2023–2025 | **Universität Münster**; Homotopy and Arithmetic working groups: $K$-Theory of the Integers · Euler Characteristics and Zeta Functions · $K$-Theoretic Poitou–Tate Duality · Linearity in $p$-adic Lie Theory · Duality and Linearization for $p$-adic Lie Groups (after Clausen) |
| 2018–2022 | **Northwestern University**; Factorization Homology · Geometry for Prime Addicts (toward Scholze's proof of the weight-monodromy conjecture) · Formal Moduli Problems · TamagaWHAT (the Tamagawa conjecture for function fields) · Applications of Topos Theory · Moduli Spaces · Stop Staring and Compute! |

## SELECTED INVITED TALKS

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| Nov 2026 | **The Lubin–Tate Action at $h = p^{k-1}(p-1)$ from a Schubert Cell of an Affine Grassmannian**; NRW Topology Seminar |
| 2026 | **Mod $p$ Langlands and the Chromatic Splitting Conjecture**; Essen Number Theory Seminar |
| 2024 | **Moduli Stacks of Curves in Homotopy Theory**; Johns Hopkins Topology Seminar |
| 2021 | **Galois Theory of Spheres**; Front Range Number Theory |
| 2020 | **Explicit Arithmetic of Jacobians**; ICERM Workshop on Arithmetic Geometry, Number Theory, and Computation |
| 2019 | **Inverse Galois Theory in the Homotopy Groups of Spheres**; Princeton CCR Colloquium |
| 2019 | **Galois Theory and Galois Representations in Homotopy Theory**; Higher School of Economics, Moscow |
| 2019 | **An Introduction to the Classic Theory of $p$-Divisible Groups**; Oberwolfach Arbeitsgemeinschaft |

## RESEARCH OUTSIDE MATHEMATICS

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| 2026 | **Memantine as a candidate non-opioid analgesic in central sensitization**; NMDA receptor antagonism · opioid tolerance and hyperalgesia · drug repurposing · trial design |
| 2026 | **Neostigmine in feline idiopathic megacolon**; enteric motility and cholinergic transmission · documentation of an unpublished treatment |
| 2026 | **One Fault, Many Names**; follow-up to Biome · habituation as a slope across body systems |
| 2024– | **Biome of Overstimulation: Sensory Sensitivity as a Unifying Mechanism**; joint with L. Estinto; sensory processing as an organizing principle · gut microbiome · autoimmunity · chronic pain · glutamate excitability |
| 2015 | **Dependent and directed type systems for multiphysics models**; Santa Fe Institute; multiscale modeling · compositional model building · topology applied to complex biological systems |
| 2014 | **Decoder convergence and optical recording for intracortical motor prostheses**; intracortical brain–computer interfaces · foreign body response and chronic signal loss |
| 2013–2015 | **Modular robotic retrofits for powered wheelchairs**; Thiel Fellowship; automated pressure redistribution · assisted transfer · gaze-controlled assistive mobility · user-led requirements |
| 2013–2015 | **Unsupervised classification of rodent ultrasonic vocalization for continuous preclinical monitoring**; Vium; computational bioacoustics · preclinical safety pharmacology · affective-state readout |
| 2013 | **Unsupervised grammar and lexicon induction from partially parallel text**; George Mason University; decipherment of incompletely understood scripts · contracted Braille as the test language |
| 2013 | **Automated abstractive summarization of biomedical literature for clinical readers**; George Mason University; computational semantics · evidence fidelity |
| 2012–2013 | **Electronic-structure prediction of superconducting transition temperature**; George Mason University, with D. Papaconstantopoulos; augmented plane wave method · intermetallic compounds · computational materials screening |
| 2012 | **Graceful degradation under sensor failure: contact-rich manipulation on proprioception alone**; George Washington University; reinforcement learning · Willow Garage PR2 · human–robot interaction |
| 2012 | **Monocular visual SLAM with motion planning on a micro aerial vehicle**; George Mason University; loop closure · scale from sonar and inertial fusion · motion planning on a partial map |
| 2012 | **Standing-wave mode structure of an RF discharge in a polyhedral cavity**; 2.45 GHz argon discharge · deployable structures in ionospheric plasma |
| 2012 | **Flexible $TiO_2$ resistive-switching devices**; Mary Baldwin College; memristive device fabrication · bend-cycle characterization · in-memory computing |
| 2011–2014 | **Portable detection of gluten and common allergenic proteins**; laser absorption spectroscopy · antibody colorimetric assay · immunoassay cost and sampling limits |

## CREATIVE WORKS

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| 2026 | ***Impaction***, first play; staged in Los Angeles as part of *Nothing to See Here*, directed by Jacques Manjarrez |
| 2025 | ***Endomortis***, comic; with Petra Flurin; crayolapunk surrealist comic about grief |